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Motivational Bases of Choice in Experimental Games[1]

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The present study presents a motivational theory of choice behavior in game situations, and develops a method that provides a flexible means for comparing the relative dominance of the various social motives postulated to underlie choice behavior in such situations. An experiment is conducted to test a set of hypothesized relationships between two conditions of interpersonal set, "partner" and "opponent"; three conditions of cumulative display of scores, "own," "joint," and "difference"; six classes of decomposed games; and the subjects' actual choice behavior. It was observed that whereas set did not affect the subject's choice behavior, there was a highly significant interaction between the display conditions and the classes of games. Finally, the relative utility of an algebraic and a stochastic choice model were evaluated in terms of the data obtained in the study.

Early investigations of human behavior in experimental games were aimed primarily at the evaluation of the normative prescriptions of formal game theory. This emphasis is evident in the thorough review of these studies by Rapoport and Orwant (1962). More recent investigations of human game behavior, however, have been less concerned with measuring the extent to which formal prescriptions describe actual choices, and have focused instead on determining the variables that influence these choices in situations of social interdependence. This is certainly true of research on cooperation and competition in Prisoner's Dilemma and related games in which meaningful normative theories are scarce. The results of these investigations have led researchers away from prescriptive economic theories of choice behavior in the direction of at least recognizing the need for psychological theories.

One major theoretical question that has received attention recently concerns the motivational orientations of subjects playing experimental games—specifically, what are their goals, and how do various experimental manipulations affect these goals? While early studies tended to assume that subjects were attempting to maximize the payoffs as defined by the experi-

[1] A brief version of this paper was given by David Messick at the meeting of the Psychonomic Society in St. Louis, October 1966. The research reported herein was supported by NSF grant GS-1032 to the authors.

menter and displayed in the payoff matrix, recent research has clearly demonstrated that this is not always the case. McClintock and McNeel (1966a) have employed a game which is similar to the Prisoner's Dilemma paradigm, but which differs in the essential feature that if subjects attempt to maximize the experimenter-defined payoffs then the choice of a particular strategy is clearly dictated, since if both players do so the payoff to each is maximized. The Prisoner's Dilemma game, of course, does not have this property and it is this fact that obfuscates conclusions regarding the motivational bases of choice in experiments employing PD games. The results of studies conducted by McClintock and McNeel (1966b, 1966c, 1966d, 1967) strongly suggest that individuals do not exclusively endeavor to maximize the experimenter-defined "own" payoffs, but are, in fact, more concerned with their scores relative to the other player than with the absolute magnitude of their own scores. Another series of experiments, reported by Messick and Thorngate (1967), also demonstrates that subjects do, in fact, tend to maximize relative gain, i.e., the *difference* between their payoffs and those of the other player, when knowledge of the other person's payoffs is available. They found that this effect is mediated more by a tendency to avoid getting less than the other player than by a desire to surpass him.

In an effort to provide a conceptual system within which the results of these and other studies can be interpreted, McClintock and Messick (1965) described a means of viewing experimental games in terms of the extent to which the payoff structure in a two-person, two-choice game enables one to infer the motives that underlie the available choices. They assume that subjects have motivational orientations to maximize the sum of the payoffs to both players (joint gain), the difference between their gain and that of the other (relative gain), and their own payoffs (own gain). These goals can be identified with cooperation, competition, and individualism, respectively, and it is assumed that the relative strengths of these orientations depend on the particulars of the experimental context.

In a typical Prisoner's Dilemma game, such as the one given below, it is seen that *given* any one of the three motives described above, one of the two choices is clearly preferable to the other in that one of the two choices dominates (in the game-theoretic sense) the other. In order to clarify this point, consider the following PDG matrix from the standpoint of Player 1 whose own payoffs are the left-hand entry in each cell.

$$\begin{array}{ccc} & & \text{Player 2} \\ & & \begin{array}{cc} B_1 & \quad B_2 \end{array} \\ \text{Player 1} & \begin{array}{c} A_1 \\ A_2 \end{array} & \begin{bmatrix} 9,9 & -10,10 \\ 10,-10 & -1,-1 \end{bmatrix} \end{array}$$

If we assume that Player 1 is attempting to maximize joint gain, the matrix can be summarized as:

$$\text{Player 1} \quad \begin{array}{cc} & \text{Player 2} \\ & \begin{array}{cc} B_1 & B_2 \end{array} \\ \begin{array}{c} A_1 \\ A_2 \end{array} & \begin{bmatrix} 18 & 0 \\ 0 & -2 \end{bmatrix} \end{array}$$

Given this matrix, the A_1 strategy clearly dominates A_2. If, on the other hand, we assume that Player 1 is attempting to maximize own gain regardless of the outcome of the game to the other player, the matrix can be summarized as follows:

$$\text{Player 1} \quad \begin{array}{cc} & \text{Player 2} \\ & \begin{array}{cc} B_1 & B_2 \end{array} \\ \begin{array}{c} A_1 \\ A_2 \end{array} & \begin{bmatrix} 9 & -10 \\ 10 & -1 \end{bmatrix} \end{array}$$

In this instance one can observe that given Player 2's choice of B_1, Player 1 should prefer A_2, and given Player 2's choice of B_2, Player 1 should again prefer A_2. Thus, given the motive to maximize own gain (and minimize own potential losses), A_2 is the preferred response.

Finally, if Player 1 is concerned with maximizing the difference between his own and the other player's score, then the matrix can be summarized as follows:

$$\text{Player 1} \quad \begin{array}{cc} & \text{Player 2} \\ & \begin{array}{cc} B_1 & B_2 \end{array} \\ \begin{array}{c} A_1 \\ A_2 \end{array} & \begin{bmatrix} 0 & -20 \\ +20 & 0 \end{bmatrix} \end{array}$$

Given this matrix, A_2 obviously dominates A_1.

Summarizing what we have observed so far, we note the following possible strategy choices dictated by several possible motives.

Possible Strategy choice	Game behavior	Possible motives
A_1	Cooperation	Maximize joint gain
A_2	Competition	Maximize own gain
A_2	Competition	Maximize relative gain

Double-Dominance Games

In the PDG, either of the two available choices can be dominant depending on which of the three motives is assumed operative. Any game in which this is true will be called a *double-dominance game*. It is pertinent to ask if there are other double-dominance matrices which permit the isolation or control of the three motives that have been assumed to be operative in the play of experimental games. Let us consider two distinct subclasses of double dominance (which we will henceforth abbreviate as D^2) games.

The first class of games is characterized by the fact that one choice (out of the two possible) dominates the other with respect to a different motive, and the third motive is nondiscriminating or constant for both strategies. (We will consider only the three motives of maximizing own gain [*o*], joint gain [*j*], and relative gain [*r*].) We abbreviate such games as $D^2_{j \cdot o(r)}$, for example, to indicate that A_1 dominates A_2 with respect to joint gain (j), A_2 dominates A_1 with respect to own gain (o), and relative gain (r) is the same for both A_1 and A_2.

Obviously there are three types (disregarding the labels of the strategies) of D^2 games in this class, $D^2_{j \cdot o(r)}$, $D^2_{o \cdot r(j)}$, and $D^2_{r \cdot j(o)}$. The first two types cannot be realized in a two-choice game, however. Representing the general game in the following general form, we see that the requirements for

		Player 2	
		B_1	B_2
Player 1	A_1	a_{11}, b_{11}	a_{12}, b_{12}
	A_2	a_{21}, b_{21}	a_{22}, b_{22}

$D^2_{j \cdot o(r)}$ are (for the row player),

$$(1)\quad a_{11} + b_{11} \geq a_{21} + b_{21}$$
$$a_{12} + b_{12} \geq a_{22} + b_{22}$$

with the strict inequality holding for at least one of the two above expressions;[2]

$$(2)\quad a_{21} \geq a_{11} \text{ and } a_{22} \geq a_{12}; \text{ and}$$

$$(3)\quad a_{11} - b_{11} = a_{21} - b_{21}$$
$$a_{12} - b_{12} = a_{22} - b_{22}.$$

Impossibility proof. Let $a_{11} - b_{11} = a_{21} - b_{21} = d_1$. If $a_{21} > a_{11}$ then $b_{21} = a_{21} + d_1 > b_{11} = a_{11} + d_1$. Thus $a_{21} + b_{21} > a_{11} + b_{11}$, which contradicts requirement (1).

A similar analysis indicates that the $D^2_{o \cdot r(j)}$ game is impossible.

The $D^2_{j \cdot r(o)}$ game, however, can be constructed. The two games below represent symmetric and non-symmetric examples of $D^2_{j \cdot r(o)}$ games, respectively:

[2] This will be assumed for all weak inequalities to be used.

		Player 2		
		B_1	B_2	
Player 1	A_1	5,5	0,5	symmetric
	A_2	5,0	0,0	

		Player 2		
Player 1	A_1	3,2	1,2	non-symmetric
	A_2	3,1	1,1	

One of the requirements of this type of game is that $a_{11} = a_{21}$ and $a_{12} = a_{22}$. This condition implies that $D^2_{j.r(o)}$ games will always be cases of "mutual fate control," in the terminology of Thibaut and Kelley (1959). That is, one's outcome will always be a function solely of the choice of the other person in the game.

In the second general subclass of D^2 games, rather than holding one of the motives constant, one motive is pitted against the other two. The choice A_1, for example, might lead to the maximization of joint gain, while A_2 might maximize both own gain and relative gain. Again neglecting the labeling of strategies, there are three types of games in this subclass. These will be called $D^2_{oj.r}$, $D^2_{or.j}$, and $D^2_{rj.o}$ to indicate that relative gain, joint gain, and own gain respectively are pitted against the other two motives.

Of the three types of games in this group, only the $D^2_{rj.o}$ class is empty. This is easily shown. The dominance requirements for $D^2_{rj.o}$ are

(1) $a_{21} \geq a_{11}$ and $a_{22} \geq a_{12}$;

(2) $a_{11} + b_{11} \geq a_{21} + b_{21}$ and $a_{12} + b_{12} \geq a_{22} + b_{22}$; and

(3) $a_{11} - b_{11} \geq a_{21} - b_{21}$ and $a_{12} - b_{12} \geq a_{22} - b_{22}$.

Impossibility proof. Let $a_{21} > a_{11}$ and $a_{11} + b_{11} \geq a_{21} + b_{21}$. Then $b_{11} > b_{21}$ which implies $a_{21} - b_{21} > a_{11} - b_{11}$, which contradicts (3).

The other two types of games are possible, however, and both have been employed in empirical research. McClintock and McNeel's (1966a) maximizing-difference game is an example of a $D^2_{oj.r}$ game, while the standard PDG exemplifies the $D^2_{or.j}$ game. Thus, of the six types of D^2 games, three are possible to instrument: (a) that game in which own gain is held constant and joint gain is compared to relative gain; (b) that game in which own and joint gain are pitted against relative gain; and (c) that game in which joint gain is pitted against own and relative gain.

In each of these three types of games a motivational conflict is established, in that the choices that satisfy one of the three motives will be opposed to the choices that satisfy at least one of the others. There are three additional games having the property that one of the two possible choices dominates the other in terms of some subset of the three motives we have been discussing, while the other choice dominates the first with respect to none. The first such game that can be constructed is one in which all three motives lead to the same choice. Such games will be denoted D_{ojr} to indicate that all three motives dictate the same choice. In the second game of this type, own gain and relative gain lead to the same choice while the two choices are equated for joint gain. Games falling into this category will be called $D_{or(j)}$ games. The final game of this type finds own gain and joint gain leading to one choice, with the choices equated in terms of relative gain. These games will be denoted $D_{oj(r)}$games.

While it may seem that these three games are of less psychological interest than the double-dominance games, since the latter involve motivational conflict while the former do not, it will become evident in discussing the results of the experiment to be described below that these games are of critical importance for the evaluation of the conceptual approach presented above, and for the testing of specific models of choice behavior.

Decomposed Games

Research studies in the area of games have traditionally displayed to subjects the payoffs associated with their choice options in the form of the payoff matrices which have been given above as examples of the various games. This particular form for displaying information was originally adopted by those investigators who were concerned with evaluating the prescriptive power of formal game theory. That is, in order to evaluate the formal theory it was necessary, insofar as possible, that the experimental circumstances conform to the assumptions of the theory since the theoretical prescriptions depend critically upon the information made available to the decision makers. Specifically, the theory assumes that each decision maker knows his own payoffs and those of the other decision maker(s) and, in addition, that he knows how these payoffs relate to the choices available to all parties involved. Inasmuch as the most efficient means of displaying this information is to present the decision maker with the payoff matrix, it is not surprising that early studies of choice strategies in experimental games almost exclusively display the payoffs of experimental subjects, and that they do so with a payoff matrix.

Although there have been numerous conceptual departures from the theoretical considerations that gave rise to game research, comparable changes have not been evident in the experimental methods employed to study this behavior. The traditional method of displaying the interdependence of outcomes in the form of a payoff matrix is not dictated by any psychological theory and would seem to be widely in use either because there are no alternative procedures for displaying such information or because of historical inertia. Since alternative procedures do exist, it becomes an important problem to determine which are the most useful in generating answers to specific theoretical questions concerning the determinants of behavior in experimental games.

In the present study, a method of displaying payoff information was devised which provides a simple, direct, and flexible means for the assessment of motives in a game-type situation, but which, in addition, preserves the property of payoff interdependence that characterizes game experimentation. This procedure, which uses what will be called the decomposed game,[3] may be described as follows. Each subject in a dyad is given a choice between two options, X^1 and Y^1 for Player 1 and X^2 and Y^2 for Player 2. Each option is an ordered pair of numbers (c,d), where the first number denotes the payoff to the subject making the choice, and the second, d, denotes the payoff to the other person. We will be concerned only with symmetric decompositions in which $X^1 = X^2$ and $Y^1 = Y^2$. Under these circumstances the correspondence between the decomposed game and the traditional payoff matrix is easily seen: if $X^1 = X^2 = (c_1, d_1)$ and $Y^1 = Y^2 = (c_2, d_2)$, then $a_{11} = b_{11} = c_1 + d_1$; $a_{12} = b_{21} = c_1 + d_2$; $a_{21} = b_{12} = c_2 + d_1$; and $a_{22} = b_{22} = c_2 + d_2$. Clearly, one can find the payoff matrix for any symmetric decomposed game.

The converse, however, is not true. It can be shown that the necessary and sufficient condition for a symmetric, two-person, two-choice game to be symmetrically decomposable is $a_{11} - a_{12} = a_{21} - a_{22}$. Furthermore, for any symmetrically decomposable game one can find an infinite number of symmetric decompositions since if $\langle X,Y \rangle$ is a decomposition, so is $\langle X^*,Y^* \rangle$, where $c_i^* = c_i + e$, $d_i^* = d_i - e$, and e is any real number.

It remains now to spell out the relationship between decomposed games and the dominance games discussed earlier. Dominance games were defined as payoff matrices having specified types of dominance relations on their rows and columns. Decomposed games can also be characterized by such dominance structures in the following manner: we say that one choice, say X, dominates the other with respect to own gain iff $c_1 > c_2$; X dominates Y with respect to relative gain iff $c_1 - d_1 > c_2 - d_2$; and X dominates Y with respect to joint gain iff $c_1 + d_1 > c_2 + d_2$. Of course the choices may be equal on any of these three value dimensions.

There are three direct consequences of this classification of decomposed games according to their dominance properties. First, excluding the null case in which the options are equal on all three dimensions, there are exactly six dominance structures that can characterize a decomposed game

[3] The decomposed game was independently devised by Pruitt (1965) within the context of Prisoner's Dilemma research.

and these are the same six that were found previously for payoff matrices. Second, unlike payoff matrices, every symmetric decomposed game is a dominance game in that it has one of six possible dominance structures. While many payoff matrices lack motivational dominance structures of the type being considered here, all decomposed games have such a structure. Finally, a decomposed game possesses a particular dominance structure, e.g., $D^2_{oj \cdot r}$ or D_{orj}, if and only if its associated payoff matrix is characterized by the same structure. This simply means that the dominance structure of the payoff matrix in a two-person, two-choice, mixed-motive game is invariant under decomposition and "recomposition." This last feature is obviously the most important of all since it guarantees a formal isomorphism of payoffs between decomposed games and payoff matrices.

Examples of two decomposed games are given below with their associated payoff matrices. It can be seen that in the decomposed game the subject has complete control over the payoffs that he and the other player will receive as a function of the subject's choice. Interdependence is maintained by informing the subjects that the payoffs to themselves and the other player are a function both of what they assign to themselves and the other player, and of the assignment made by the other player.

Decomposed D_{ojr} Game

		Choices	
		X	Y
Payoffs	Own	3	1
	Other	1	0

Corresponding Matrix Game

		Player 2	
		X	Y
Player 1	X	4,4	3,2
	Y	2,3	1,1

Decomposed $D^2_{oj \cdot r}$ Game

		Choices	
		X	Y
Payoffs	Own	5	8
	Other	2	6

Corresponding Matrix Game

		Player 2	
		X	Y
Player 1	X	7,7	11,10
	Y	10,11	14,14

The motivational isomorphism between decomposed and matrix games is nicely illustrated by these examples. In the second example, the $D^2_{oj \cdot r}$ game, it is clear the subject will choose Y either if he is trying to maximize his own score $(8 > 5)$ or if he is trying to maximize the joint score to the dyad $(8 + 6 > 5 + 2)$. He will choose X, however, if he is trying to maximize relative gain $(5 - 2 > 8 - 6)$. Inspection of the corresponding matrix game reveals that the Y choice dominates the X choice with respect to own $(10 > 7$ and $14 > 11)$ and joint $(10 + 11 > 7 + 7$ and $14 + 14 > 11 + 10)$ gain, while X dominates Y with respect to relative gain

($7 - 7 > 10 - 11$ and $11 - 10 > 14 - 14$). Thus, the motivational implications of the two choices are equivalent in the decomposed and the matrix games.

From the above examples, it is apparent that the experimenter has considerable flexibility in changing the motivational structure of the situation for subjects through the course of an experiment. He can manipulate not only the relationships among the three motives under consideration, but can change their relative magnitudes by systematically manipulating the absolute and relative point values afforded the subject and the other player. This capability may provide a possible means of scaling the relative strengths of the various motivational predispositions that are represented in decomposed games.

Finally, it should be noted that decomposed games provide a particularly useful method for measuring interpersonal motives. Research to date in the area of games fails to differentiate clearly between motivation (goals) and strategy (instrumental acts). For example, in a typical study a subject may adopt a cooperative strategy (permitting both own and other to be rewarded) in order to satisfy one of several different motives. In the present task the structure of the choice matrix is designed to differentiate between motivational orientations, and to afford the subject little control over the other player's behavior. Given this, the preferences of the subject would seem more likely to be a direct expression of his goals rather than a method for influencing the other player's choices as an indirect method for achieving his goals. We do not mean to imply that influencing the other's behavior is an unimportant consideration in game behavior. But it seems more appropriate to first find a way of assessing the subjects' goals before attempting to determine the strategies they adopt to achieve these goals.

METHOD

The purpose of the present study was to determine the differential effects of instructional set and display of cumulative-score information upon the choice behavior of subjects confronted with each of the types of decomposed games described previously. Specifically, the study crossed two types of interpersonal set with three types of cumulative-score feedback, thus generating six experimental conditions. Subjects in each of the six conditions played the same sequence of decomposed games. The two conditions of interpersonal set were established by characterizing the other player as either "partner" or "opponent." This manipulation was effected in two ways. First, in the instructions the other person was referred to as "partner" for half the dyads and as "opponent" for the other half. Second, for those subjects who received the "partner" instructions the displays on which the other person's scores were exhibited were labeled PARTNER'S OUTCOME. The other half of the subjects had these displays labeled OPPONENT'S OUTCOME. These labels were present and visible to the subjects throughout the experiment. The three score-feedback manipulations were established by: (a) displaying only own cumulative score; (b) displaying the joint cumulative score of both players; and (c) displaying the difference between own and other's cumulative scores. It was anticipated that the set variation, to the extent that it influenced the motivational predispositions of the players, would produce more joint-gain choices under the "partner" induction and more relative-gain selections under the "opponent" induction. The feedback induction, a variation upon one used in an earlier study by McClintock and McNeel (1966d) was assumed to be the stronger of the two manipulations. It was hypothesized that the display of the cumulative differences between the players' scores would produce markedly more relative-gain preferences than the other two display conditions. Between the own- and joint-score displays, it was anticipated that those receiving own scores would make more own-gain responses, whereas those receiving the joint-score display would make more joint-gain choices.

Procedure

Subjects for the study were 180 female students, 30 in each of the six experimental groups, drawn from introductory psychology courses. They were randomly assigned to dyads subject to the constraint that they were relative strangers, the latter being defined in terms of their not knowing the other player well. They were seated on opposite sides of a table by an experimenter and were visually isolated from one another by a partition. Mounted on this partition were a set of digital display units by means of which the point matrices for each game were displayed, and by which cumulative scores were displayed. In the point matrices the two choices were identified as column A or column B. The subjects were provided with two silent switches corresponding to the two columns by which they indicated their choice for each game.

Identical instructions were given to all subjects by tape recorder except for those changes necessitated by the major experimental manipulations. The instructions noted that the subjects' total scores at the end of the game would be a function both of the points they awarded themselves and of those afforded them by their "partner" (or "opponent"). They were further informed that they would receive a small monetary reward which would vary as a function of their *own* total score at the end of the game. As part of the instructions, all subjects played several practice games to familiarize themselves with the task and the meaning of the displays.

Subjects were then presented with 80 different two-choice matrices. The six game structures noted previously were represented as follows: 30 $D^2_{oj.r}$ games; 16 $D^2_{j.r(o)}$; 6 $D^2_{or.j}$; 16 $D_{oj(r)}$; 6 D_{ojr}; and 6 $D_{or(j)}$. Members of the various game structures were presented in a random order.

After the subjects had made their choices on each trial, their cumulative-score boards were updated to reflect one of the three feedback conditions, and then they were presented with the subsequent choice matrix. Upon completion of the 80 trials, subjects received nominal payments as a function of their total scores (maximum of $1.25) and then signed a statement that they would not discuss the study until after a specified date.

RESULTS

To prepare the data for analysis and to permit comparison of choices across the six classes of matrices, subjects' 80 choices were coded in the following manner: an *A* or *B* choice was assigned the value "1" if the subject chose that option having the largest own gain or, for $D^2_{r.j(o)}$ games in which own gain is identical for both choices, if he chose that option having the largest relative gain. Otherwise, the choice was scored "0." Thus, for five classes of games (the exception being the $D^2_{r.j(o)}$ class) the choice proportions that will be analyzed represent the proportion of choices in favor of own-gain optimization. In the case of $D^2_{r.j(o)}$ games, where own gain is the same for either choice, the reported proportions represent the proportion of relative-gain maximization choices across the 16 game matrices falling in this class.

Using this coding scheme, choice proportions were determined for each of the six types of games in each of the six experimental groups. An analysis of variance was then performed on these proportions to measure the effects of the experimental treatments and the motivational structure of the games on subjects' choices. This analysis is summarized in Table 1.

TABLE 1
SUMMARY OF ANALYSIS OF VARIANCE

	MS	*df*	*F*
Between			
Set	.00	1	.00
Display	.02	2	.40
S × D	.01	2	.20
Error_b	.05	84	—
Within			
Games	2.10	5	70.00*
G × S	.06	5	2.00
G × D	.57	10	19.00*
G × S × D	.05	10	1.67
Error_w	.03	420	—

* $p < .01$.

As was anticipated, the analysis shows no overall effects across the six game classes for instructional set or feedback display. However, a large effect is present both for classes of games, indicating that subjects' choices depend upon the motivational structure of the various classes of games, and for the interaction of games with feedback display. This significant interaction indicates that the effects of the feedback manipulation depend upon the motivational structure of the class of game being played. The expected set ("partner" vs. "opponent") by games interaction failed to reach acceptable levels of significance. The data were therefore combined across the set conditions for subsequent analyses.

The mean proportions for the three display conditions and the six game structures are presented in Table 2. In this table the types of games are identified by the ordered 3-tuples heading each row. The 3-tuples are ordered from left to right in terms of own, relative, and joint gain. A "1"

TABLE 2
CHOICE PROPORTIONS FOR SIX GAME TYPES IN THE THREE EXPERIMENTAL CONDITIONS

Condition		Display: Own	Display: Difference	Display: Joint
D_{orj}	111	.910	.885	.950
$D_{or(j)}$	11X	.865	.915	.865
$D_{oj(r)}$	1X1	.840	.680	.865
$D^2_{or.j}$	110	.785	.860	.750
$D^2_{oj.r}$	101	.730	.350	.685
$D^2_{r.j(o)}$	X10[a]	.600	.895	.555

[a] The proportions in this row represent the proportions of relative-gain choices.

is entered in the first position to indicate that an own-gain choice is scored "1," unless the two options are equal in own gain in which case an "X" is entered in the first position. A "1" is placed in the second position if relative gain leads to the same choice as own gain. If relative gain leads to the other choice, the second position is filled with a "0"; if relative gain is constant, an "X" is entered. The third position is filled in the same way: a "1" is entered if joint gain leads to the same choice as own gain, etc.

One interesting aspect of the obtained proportions is the similarity between the choices of the Own-Display group and the Joint-Display group. It appears that Own and Joint feedback lead to similar motivational orientations in subjects. The Difference-Display group, however, is markedly different from these two groups. With the $D^2_{j.r(o)}$ games, for example, in which own gain is equal for the two choices and relative gain leads to a different choice from joint gain, the Own- and Joint-Feedback groups chose the relative-gain maximization alternative 60 and 56% of the time, respectively, while the Difference group chose that alternative nearly 90% of the time. The difference between this latter group and the former two is even more apparent within the $D^2_{oj.r}$ games. The proportions of own- and joint-gain choices for the first two display groups are .730 and .685, respectively, while for the Difference-Feedback group it is .350.

The strong interaction between Score Feedback and Game Structure suggests that care must be taken in the interpretation of research in which effects found using one type of game are (or are not) replicated using another type of game. There are currently a number of types of games being used in experimental studies. There are, for example, the well-known PDG, the "Maximizing Difference" game used by McClintock and McNeel (1966a), the "Chicken" game employed by Sermat and Gregovitch (1966), and the game employed by Messick and Thorngate (1967). While the latter two games are not dominance games of the sort being considered here, the implications of results of the present study apply equally well to them—the point being that in order to predict the effects of a given experimental manipulation on the choice behavior of subjects playing a game, it is necessary to consider carefully the motivational structure of the game itself. This point was documented from a somewhat different perspective by Messick and Thorngate (1967).

It is possible to obtain additional insight into the relative strength of the three motives in the present experimental situation by comparing the behavior of subjects within particular classes of games. For example, if it is the case that joint-gain maximization is a weaker tendency than relative-gain maximization, then the presence or absence of joint-gain choices in the motivational structure of the games should have less effect on choice behavior than the presence or absence of relative-gain choices. There should be less of an effect on the choice proportions if joint gain is pitted against own and relative gain than if relative gain is pitted against own and joint.

It is possible to evaluate the contributions of joint and relative gain by making joint (relative) gain equal for both possible choices and comparing the resulting choice proportions to those which obtain when all three motives dictate the same choice. This involves a comparison of D_{ojr} games with $D_{or(j)}$ and $D_{oj(r)}$ games. To the extent that joint-gain (relative-gain) maximization is involved in the D_{ojr} games, the difference will provide a measure of the magnitude of the involvement. Furthermore, comparing the $D^2_{or\cdot j}$ game, in which joint gain is no longer equated for the two choices but is pitted against own and relative gain, with the D_{ojr} and $D_{or(j)}$ games, will provide a further measure of the strength of the joint-gain tendencies; comparing the $D^2_{oj\cdot r}$ with the D_{ojr} and $D_{oj(r)}$ games will provide a measure of the strength of the relative-gain tendencies. As one goes from D_{ojr} games to $D_{or(j)}$ games, and to $D^2_{or\cdot j}$ games, the proportions of own- and relative-gain choices should decrease, and the rate of decrease should be a measure of the extent to which joint-gain maximization is a basis for choice. And as one goes from D_{ojr} to $D_{oj(r)}$ to $D^2_{oj\cdot r}$, the proportion of own- and joint-gain choices should decrease, and the rate should measure the degree to which relative gain is a basis for choice. (These ideas will be given mathematical expression presently in terms of a stochastic model of the motivational determinants of choice in such contexts.)

The relevant proportions are plotted in Fig. 1a for joint gain and in Fig. 1b for relative gain for each of the display conditions. As one can see in these figures, in all cases but one there is a decrease in the choice proportions as one goes from D_{ojr} to $D_{or(j)}$ or $D_{oj(r)}$ games, and as one goes from $D_{or(j)}$ and $D_{oj(r)}$ to $D^2_{or\cdot j}$ and $D^2_{oj\cdot r}$ games, respectively. Furthermore, for all three experimental groups, the rate at which the proportions decline across the three relevant game classes is greater for relative than for joint gain. As might be expected, this difference in rate is most pronounced in the Difference-Feedback groups. Thus, these data tend to support, although not as strongly perhaps as might be desired, the hypothesis that relative-gain maximization is a more prominent motive than joint-gain maximization.

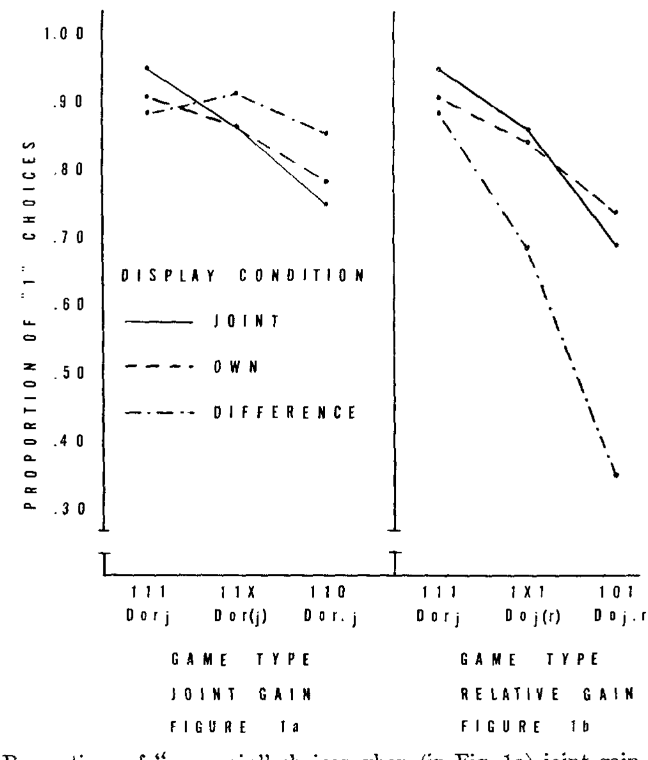

FIG. 1. Proportions of "own gain" choices when (in Fig. 1a) joint gain leads to the same choice (D_{orj}), when joint gain is nondiscriminating ($D_{or(j)}$), and when joint gain opposes own gain ($D^2_{or.j}$). Figure 1b presents the same sequence of contrasts for relative gain.

There is one final and direct test which can be made of the relative strength of joint- vs. relative-gain motives. In the $D^2_{j.r(o)}$ game, the two choices are equated in terms of own gain. Thus subjects will choose on the basis of either joint gain or relative gain. Keeping in mind that relative-gain choices in these games are scored "1" and joint-gain choices "0," the choice proportions for the $D^2_{r.j(o)}$ game in Table 2 measure the relative dominance of relative-gain choices over joint-gain choices in the three experimental groups. If relative-gain maximization is a stronger motive than joint-gain maximization, these proportions should be greater than .50. This is in fact the case for all feedback conditions as can be seen in the last row of Table 2. Again the Difference-Feedback group shows the strongest dominance of relative-gain maximization within the three experimental groups.

So far this section has been confined simply to a description of the results of the experiment. The remainder will be devoted to an attempt to provide a coherent organization for these data in terms of a mathematical model of the motivational bases of choice. Two such models will be evaluated.

An Algebraic Choice Model

Messick and Thorngate (1967) presented data which demonstrated that the value associated with a joint payoff (a,b), where a is subject's payoff and b is payoff to the other, is a function of the quantity $(a - b)$ as well as of a. They suggested that a feasible representation of the determinants of the attractiveness or reinforcingness of such outcomes might be embodied by the following model:

$$v(a,b) = f(a) + g(a - b), \tag{1}$$

where v is a utility or value measure and f and g are unknown functions describing the contributions of own gain and relative gain, respectively. The only assumption regarding f and g that would seem to be necessary is that f be monotone increasing. A number of interesting questions then arise concerning g. For example, is g monotone increasing? This would seem to be implied by theories such as those of Thibaut and Kelley (1959) or Patchen (1961) relating satisfaction with social outcomes to interpersonal comparisons of outcomes. On the other hand, Homans' (1961) concept of "distributive justice" might be taken to mean that g is unimodal, taking a maximum at 0, for example, if the two people are of equal perceived status. To evaluate this model in the context of the present experiment, it is assumed that one choice, $X = (a_1,b_1)$, is preferred to another, $Y = (a_2,b_2)$, if and only if $v(a_1,b_1) > v(a_2,b_2)$ or, in terms of the model,

$$f(a_1) + g(a_1 - b_1) > f(a_2) + g(a_2 - b_2). \tag{2}$$

Joint gain is ignored in this model both to make the model mathematically tractable and also as a result of the data previously reviewed which indicate that relative-gain maximization is a more important choice determinant than joint-gain maximization.

Any strict algebraic choice model requires transitivity of choices. While the games used in this experiment were not selected primarily to allow for a test of this model, there are two sets of three games each on which transitivity can be tested. If all subjects made their choices completely randomly, one would expect 25% of the observed choices in these triples to be intransitive. Of the 360 tests of transitivity made (2 triples for 180 subjects), 27 intransitive triples were found. This is much less than the 90 expected if choices were made randomly. However, the conclusion is forced that *no* algebraic model will handle the choices of 26 (one subject had two intransitivities) of the subjects.

To further examine the suitability of the algebraic model a test was made of the *consistency* of the subjects' choices. The test was constructed as follows: five pairs of games were selected such that for each pair the choice of one of the two options must be made solely on the basis of the function g, and for each game in the pair the choice involves the same arguments of g. One such pair of games, for example, is given below:

Game 22		Game 54	
X	(8,0)	X	(9,1)
Y	(8,7)	Y	(9,8)

In Game 22 a choice of X, according to the model under consideration, implies $v(8,0) > v(8,7)$ or $f(8) + g(8) > f(8) + g(1)$. This, of course, implies that $g(8) > g(0)$. In Game 54, an X choice has the identical implication. Thus, if this algebraic model is consistent, an X choice in Game 22 should be accompanied by an X choice in Game 54. Otherwise, it is necessary to assume either that the model does not hold, or that the functions f and/or g change over time. In either case the model would be of little value in the interpretation of the results of this experiment.

Of the 154 individuals who satisfied the transitivity requirement, 105 made at least one pair of inconsistent choices out of the five possible. Thus, only 49 of the original 180 subjects in the experiment satisfy both transitivity and consistency. These results raise serious doubts regarding the usefulness of this algebraic model.

A final test was made of the assumption that f is monotone increasing. If f is a monotone-increasing function, then for $x > y$, $f(x) > f(y)$. This property can be checked in games which differ in own gain but for which relative gain is constant, i.e., the $D_{oj(r)}$ class of games. If the choice having the larger own gain is selected in such games, according to the model

$$f(a_1) - f(a_2) > g(a_1 - b_1) - g(a_2 - b_2) = 0,$$

where $a_1 > a_2$. Sixteen of the games used in this experiment permit this test of monotonicity. Of the 49 subjects whose choices were both transitive and consistent, only 14 gave choices that were consistent with the assumption of monotonicity.

These three simple tests of the adequacy of the algebraic model, transitivity, consistency, and monotonicity of f, leave only 14 of the original 180 subjects satisfying all three. Thus the model can hardly be said to represent the underlying structure by means of which subjects generate their choices. It should be remarked that only 25 of the 80 games employed were required to violate some aspect of the model for 166 of the subjects. The failure of this model to account for the data leads to the consideration of a stochastic choice model.

A Stochastic Choice Model

The model to be described is a simple translation of the conceptual outline presented in the first section of this paper into a probabilistic choice mechanism. It is assumed that for each game, the subject is in one of four states. These states are Individualistic (s_1), Competitive (s_2), Cooperative (s_3), or Indifferent (s_4). If he is in one of the first 3 of these states, he makes his choice on the basis of own-, relative-, or joint-gain maximization, respectively. If he is in the Indifferent state, he makes his choice randomly, choosing X with a probability of 1/2. The probabilities of the subject being in each of the four states are w, x, y, and z, respectively. It is assumed that these probabilities are constant across individuals and across trials. The choices are assumed independent of the previous choices of the subject and of the other.

If, on a given trial, the subject is presented with a game that does not allow him to make a well-defined choice given his current state, it is assumed that he moves to a new state. The probabilities of moving to any state are independent of the original state and are assumed equal to w, x, y, and z, respectively. It is, then, possible for him to return to the original state. If so, he moves again. This continues either until he is in the Indifferent state or until he is in a state for which the choice for that game is well defined.

It is now possible to write the expected choice proportions for the different classes of games as a function of the parameters postulated in the model. In the D_{ojr} game, in which all three motives lead to the same choice, the subject will fail to choose the option with the largest own gain only if he is in s_4. Furthermore, given that he is in this state, he will choose the own-gain maximization alternative half the time by chance. Thus,

$$E[P(111)] = w + x + y + z/2, \tag{3}$$

and

$$1 - E[P(111)] = z/2. \tag{4}$$

Equation (3) is obtained by summing the products of $P(s_i)$ and $P(111/s_i)$

for all i. For the first three states, the conditional probabilities are 1. In other words, given that he is in any of the first three states, the probability is unity that he will make an own-gain maximization choice since all three of the criteria used for the respective states lead to the same choice. The conditional probability of an own-gain choice given s_4 is 1/2. Thus equation (3) is simply $\Sigma^4_{i=1}P(111/s_i)P(s_i)$.

Following a similar line of reasoning, the expected choice proportions for the other two games in which the first 3 motive states lead to a well-defined choice can be derived. They are given below.

$$E[P(110)] = w + x + z/2 \tag{5}$$

$$E[P(101)] = x + y + z/2 \tag{6}$$

For the three types of games in which one of the three criteria is equated for both choices, e.g., the $D_{or(j)}$ game in which joint gain is the same for both choices, the derivation of expected choice proportions as a function of the four parameters proceeds somewhat differently. To exemplify the derivations, consider the $D_{or(j)}$ game in which joint gain is constant for both alternatives. We wish to find the expression for $E[P(11X)]$. First we find the probability that the subject is in s_1 when the choice is made. There are an infinite but denumerable number of mutually exclusive and exhaustive ways in which this can occur. First, with probability w, he will be in s_1 originally and will thus make his choice without having to move to a new state. However, with probability y, he will be in s_3 and, since the alternatives are equated in terms of joint gain, a well-defined choice will not be possible and he will move to a new state. With probability w, the new state will be s_1. Thus, after one move, the probability that the subject is in state s_1 is yw. This first move may, with probability y, put him back in s_3, in which case another move will be made. Thus the probability that he will enter s_1 on the second move is y^2w. In general, the probability that he will enter s_1 after n moves is y^nw. Consequently, the probability that the subject will be in s_1 when the choice is made is

$$\begin{aligned} P(s_1) &= w + yw + y^2w + \cdots \\ &= w(1 + y + y^2 + \cdots). \end{aligned} \tag{7}$$

The expression in parentheses is the sum of a geometric series which is known to be $(1 - y)^{-1}$. Thus,

$$P(s_1) = w/(1 - y). \tag{8}$$

Likewise, for s_2 and s_4,

$$P(s_2) = x/(1 - y),$$

and

$$P(s_4) = z/(1 - y).$$

Therefore,

$$\begin{aligned} E[P(11X)] &= (w + x)/(1 - y) + z/2(1 - y) \\ &= [2(w + x) + z]/2(1 - y). \end{aligned} \tag{9}$$

Expressions for $P(1X1)$ and $P(X10)$ can be found in a similar fashion; they are given below.

$$E[P(1X1)] = [2(w + y) + z]/2(1 - x) \tag{10}$$

$$E[P(X10)] = (2x + z)/2(1 - w) \tag{11}$$

In order to evaluate this model, it is necessary to obtain estimates of the four parameters. This was done in the following way. Equation (4) was used to obtain an estimate of z. Specifically,

$$\hat{z} = 2[1 - P(111)]. \tag{12}$$

An estimate of x, $\hat{x}$, was found by the difference

$$\hat{x} = P(111) - P(101), \tag{13}$$

and for y,

$$\hat{y} = P(111) - P(110). \tag{14}$$

Finally, w was estimated by

$$w = 1 - \hat{x} - \hat{y} - \hat{z}. \tag{15}$$

Only three of the six proportions in each experimental group are used to estimate the parameters, leaving three proportions, $P(11X)$, $P(1X1)$, and $P(X10)$, on which predictions can be tested. The parameters for each of the three groups were found and are displayed in Table 3. The orderings of

TABLE 3
ESTIMATED PARAMETERS FOR STOCHASTIC CHOICE MODEL

	Display condition		
	Own	Joint	Relative
$\hat{w}$	.465	.435	.215
$\hat{x}$	.230	.265	.535
$\hat{y}$	.125	.200	.020
$\hat{z}$	.180	.100	.230

these estimates are of interest since the parameters are interpretable with relation to the experimental manipulations. In all cases, the probability of any of the first three states is largest in the experimental condition which makes that particular motivational orientation salient. The probability of s_1, Individualism, is largest in the Own-Feedback display condition; the probability of s_2, Cooperation, is largest in the Joint display condition; and the probability of s_3, Competition, is the largest in the Relative display group. While these results lend a degree of support for this model, the critical evaluation must depend on the extent to which accurate predictions can be made of the choice proportions in the three games that were not used in obtaining the parameter estimates.

The observed and predicted choice proportions for these three types of games are displayed in Table 4 for the three experimental groups. In all

TABLE 4
OBSERVED AND PREDICTED CHOICE PROPORTIONS FOR $D_{or(j)}$, $D_{oj(r)}$, AND $D^2_{r.j(o)}$ GAMES FOR THE THREE EXPERIMENTAL GROUPS

	Own		Joint		Relative	
Group	obs.	pred.	obs.	pred.	obs.	pred.
$P(11X)$	.865[a]	.897	.865[a,b]	.938	.915	.883
$P(1X1)$	.840[a]	.883	.865[a,b]	.932	.680[a,b]	.753
$P(X10)$	.600	.598	.556	.558	.895[a,b]	.828

[a] Significantly different ($p < .05$) by binomial test.
[b] Significantly different ($p < .05$) by t test.

cases, the predictions are reasonably close to the observed proportions. The largest discrepancy is .073, while the smallest is .002. In light of the fact that the procedure for estimating the parameters on which these predictions are based was not derived in such a way as to maximize the fit of the model to the data (e.g., least squares, maximum likelihood, or minimum chi-squared procedures) the predictions seem satisfactory.

The model that generated the predictions assumes that the choice probabilities are constant across subjects and for each of the three types of games. This assumption, in particular, and the other assumptions of the model imply that each observation is a Bernoulli random variable drawn from the same population and, in addition, that the sum of the observations across subjects and like-structured games is a binomial random variable. Consequently it is appropriate to test the fit of each prediction to the observed proportion by computing the normal deviate associated with the standardized difference between the observed and predicted proportions. This was done for all nine proportions using Eq. (16),

$$z = (nP - np)/(npq)^{1/2}, \tag{16}$$

where P is the observed proportion, p the predicted proportion, $q = 1 - p$, and n is the number of observations. There were 6 $D_{or(j)}$ games and 16 of each of the other two types. With 60 subjects in each of the groups, there are 360 observations for the $D_{or(j)}$ games and 960 for the other two. Using this test, six of the nine observed proportions are significantly (at the .05 level) different from the predicted values.

While the assumption that choice probabilities are constant across subjects makes for mathematical convenience, it is not an essential part of the *psychological* theory being presented here. It is, in fact, realistic to suppose that the probabilities of being in each of the four motivational states vary somewhat from individual to individual. It will still be assumed that the probabilities are constant within an individual.

If the assumption that the probabilities vary from subject to subject is correct, the binomial sampling scheme is an inappropriate model. The correct model for such a case is the *Lexis model* (see Kenney and Keeping, 1951). The difference between the two processes in terms of a physical analogy is the fact that the binomial distribution is based on the assumption that 360 (or 960) observations are drawn with replacement from an urn containing p "successes," while the Lexis scheme assumes that 60 samples of six (or 16) observations each are drawn with replacement from 60 different urns having possibly different values of p, p_i, $i = 1, 2, 3, \ldots, 60$, and in which $(1/60)\Sigma_i p_i = p'$.

One of the major differences between the binomial and the Lexis schemes is that the sampling variance for the Lexis scheme is always greater than that for the binomial. Specifically, the variance for the number of "successes" in t trials under the binomial assumptions is given by

$$Var_b(r) = tp(1 - p), \tag{17}$$

while the variance for the same quantity under the Lexis assumptions is given by

$$Var_L(r) = tp'(1 - p') + t(t - 1)Var(p_i), \tag{18}$$

where $Var(p_i)$ is the variance of the p_i. In many cases, $Var(p_i)$ is unknown, as in this study. However, it can be estimated by the following quantity:

$$est[Var(p_i)] = [Var_L(r) - Var_b(r)]/t(t - 1), \tag{19}$$

where $Var_L(r)$ is taken as the observed variance of the empirical distribution of choices.

In an effort to evaluate the binomial assumption, frequency distributions were plotted for the number of own-gain choices (relative-gain choices for the $D^2_{r.j(o)}$ games) for each type of game in each experimental group. Each of the nine distributions thus obtained was materially different from the expected binomial distribution. The statistic defined in Eq. (19) was therefore computed to determine if the observed variances were larger than expected binomial variances. The obtained estimates of the between-subject variability in choice probabilities are presented in Table 5. The first noticeable aspect of these estimates is the fact that they are all positive. This will be true if and only if the observed variances are larger than the expected binomial variances, as can be seen from Eq. (19). The second feature is an apparent interaction between game type and experimental treatments. In the $D_{or(j)}$ games, in which joint gain is constant, the Joint-Feedback group shows the greatest variance of the three groups. In the $D_{oj(r)}$ games, the Relative-Feedback group has the highest variance. Finally, in the $D^2_{r \cdot j(o)}$ games, the variance of the Own-Feedback group is second largest. This pattern of results suggests that an effect of the Feedback manipulation is to decrease intersubject variability in choice probability when the criterion appropriate to the motivational state aroused by the Feedback discriminates between the choices, but that it tends to increase intersubject variability when the criterion emphasized by the heightened motivational state is not present.

TABLE 5
ESTIMATED VARIANCES OF CHOICE PROBABILITIES FOR $D_{or(j)}$, $D_{oj(r)}$, AND $D^2_{r \cdot j(o)}$ GAMES FOR THE THREE EXPERIMENTAL GROUPS

	Display		
Condition	Own	Joint	Relative
$11X$	.017	.030	.014
$1X1$	.028	.032	.048
$X10$	.081	.111	.013

The fact that intersubject variability is not zero suggests that the binomial test used to evaluate the goodness of the fit of the predicted proportions to the observed is inappropriate, since it is based on the assumption of binomial variances. Thus, t tests were computed, employing the observed variances, to test the adequacy of the predictions.[4] Four of the nine observed proportions were significantly different from the predicted values using this procedure. Furthermore, all predictions for the Own-Feedback group are acceptable. In view of the small numerical discrep-

[4] If it is assumed that the choice probabilities differ from subject to subject, the predictions derived in equations (9) through (11) must be viewed as approximations. This results from the fact that $[2\Sigma(x_i + w_i) + \Sigma z_i]/2\Sigma(1 - y_i)$, from Equation (9), for example, is only an approximation of $(1/n)\Sigma[2(x_i + w_i) + z_i]/2(1 - y_i)$.

ancies between the observed and predicted values, and in view of the rather gross techniques used to estimate the parameters of the model, this stochastic theory of the motivational bases of choice in experimental games must, unlike the algebraic model described earlier, remain a plausibility.

DISCUSSION

The present study was conducted to determine the effects of set and of display of cumulative score upon choice behavior in decomposed games. The latter were employed in the present study primarily for the exploratory purpose of developing more effective means for isolating the social motives that underlie cooperative and competitive behavior. The formal algebraic equivalence between decomposed games and the more classically employed non-zero-sum payoff games by no means implies a psychological equivalence between the two. Pruitt (1965), in fact, found marked differences between the proportions of cooperative choices in a traditional Prisoner's Dilemma Game, and in several decompositions of the same game. This would suggest that different decompositions of a given game can change the motivational structure of that game.

The potential advantages of decomposed over the more classic matrix games are several. First, they provide a more flexible structure for representing various social outcomes. Secondly, decomposed games seem more amenable to attempts to scale and to compare the utility of various social outcomes to subjects. For example, it should be possible to obtain measures of the similarity of two social outcomes represented in decomposed games by having subjects make direct similarity judgments or by measuring their choice latencies. In addition, one could use preference data collected across a number of outcomes for purposes of scaling the relative utility of various outcomes for the subject.

In the present experiment, data were obtained which strongly suggest that the decomposed game provides a means for assessing the motivational orientations of subjects, and that these orientations can be affected, at least in part, by experimental manipulations of the sort used in the study. Thus, it was observed that the nature of the cumulative-score feedback provided to the subjects affected their choice behavior. Most significantly, those subjects receiving relative-score information played to maximize the difference between their own and the other player's scores more frequently than subjects given only own or joint cumulative scores. This finding is consistent with two previous studies in which differences have been found in choice probabilities as a function of whether one subject did or did not know the values of the payoffs received by the other. In the first of these studies, Estes (1962) found that his "scanning" model, which predicts asymptotic proportions solely on the basis of own gain, fit data of subjects who were ignorant of other's outcomes but failed to account for the data of subjects who had knowledge of the payoffs to the other. In a second study, conducted by Messick and Thorngate (1967), large differences in choice probabilities were found depending on whether other's scores were or were not displayed. These authors demonstrated, in addition, that the difference was due to relative-gain maximization by subjects who knew the other's scores.

Variations in the values of the payoffs may also be assumed to affect subjects' tendencies to cooperate, to compete, or to attend solely to own gain. While no evidence was obtained in the present study to suggest that choice probabilities differed for games *within* dominance classes, previous studies have found such relationships. (All of these studies have involved repeated plays of the same game, however, while in the present study each game was played only once.) Rapoport and Chammah (1965), for example, have shown that cooperative behavior in the Prisoner's Dilemma, a $D^2_{j \cdot or}$ game, is a function of the values of the payoffs, and McClintock and McNeel (1966a), using a $D^2_{oj \cdot r}$ game, found more own-gain (or joint-gain) choices when subjects (in this case students in Belgium) were told that each point was worth .5 francs than when they were told that each point was worth .05 francs. A similar finding is reported by Gallo (1966) in a simple bargaining context. While it is hypothesized that the behavioral effects of such variations in payoffs are mediated by changes in subjects' motivational orientations, it remains a task for future research to delineate the details of the process.

In the present study, the orientation of one player toward the other was manipulated by using the label "partner" for half of the dyads and "opponent" for the other half. This manipulation did not affect the motivational orientation of the players as hypothesized. The reasons for the failure of the manipulation are not immediately evident, although one might suspect that the manipulation was too weak to produce a systematic effect across the 80 different decomposed games employed in the present study.

Finally, the two choice models that are described above represent initial attempts to translate the conceptual approach outlined in the first section of this paper into a formal mathematical structure. An evaluation of these models suggests that this can most usefully be accomplished by the postulation of probabilistic, as opposed to completely deterministic, choice mechanisms. The probabilistic model, moreover, which predicts reasonably well the choice proportions in this study, must be viewed as a provisional statement of the *type* of motivationally rich formalization which seems to be called for by an increasing amount of data.

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